

Adaptive Hybrid Critics for Stable Value Estimation in Multi-Agent Reinforcement Learning*

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Abstract

Centralized Training with Decentralized Execution (CTDE) has become a standard paradigm in Multi-Agent Reinforcement Learning (MARL), yet choosing between a fully centralized critic (MAPPO) and a fully decentralized critic (IPPO) often introduces a rigid trade-off between bias and variance. In this work, we propose Adaptive-MAPPO, a framework that unifies these approaches by learning to dynamically interpolate between global and local value estimates. We introduce a lightweight gating mechanism that computes a per-agent mixing coefficient based on value-level features, enabling adaptive reliance on centralized information for coordination and decentralized signals for stability. We evaluate our method across diverse environments in VMAS and PettingZoo against IPPO, MAPPO, and fixed interpolation baselines. Our results show that Adaptive-MAPPO learns environment-dependent critic weighting and improves training stability in most of our environments, reducing late-training performance collapse compared to fixed-weight alternatives, although it remains less effective than the best fixed hybrid in tasks with asymmetric observations.

1. Preliminaries: BenchMARL

BenchMARL [1] is a Multi-Agent Reinforcement Learning (MARL) framework developed by Meta AI for standardized and reproducible experimentation. It provides a unified interface for training and evaluating MARL algorithms, reducing inconsistencies arising from implementation details, hyperparameters, and evaluation protocols. Built on top of `TorchRL` [2] and configured via `Hydra` [3], BenchMARL supports modular composition of environments, policies, and training pipelines. It includes implementations of common MARL algorithms such as MAPPO and IPPO, and integrates standard environments including VMAS and PettingZoo. This standardized design enables consistent benchmarking across algorithms and environments with minimal changes to the experimental setup.

2. Background: IPPO and MAPPO

2.1 Independent Proximal Policy Optimization (IPPO)

IPPO [4] applies Proximal Policy Optimization (PPO) [5] independently to each agent without any explicit coordination mechanism. Each agent learns a decentralized policy based on its own action-observation history:

$$\pi^a(u_t^a \mid \tau_t^a), \quad (1)$$

where τ_t^a denotes the local trajectory of agent a . In addition to the policy, each agent also learns a local value function $V_\phi(z_t^a)$ based on its own observations. This local critic is used to compute advantage estimates via Generalized Advantage Estimation (GAE) [6], making IPPO a fully actor-critic method applied independently per agent.

*Our code implementation is available at: https://github.com/arshiv-web/Adaptive_MAPPO_MARL

A key factor contributing to the effectiveness of IPPO is the PPO clipping objective:

$$L^{\text{CLIP}}(\theta) = \mathbb{E} \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right], \quad (2)$$

which stabilizes training by constraining policy updates. In multi-agent settings, where the environment becomes non-stationary due to concurrently learning agents, this clipping mechanism plays a crucial role in preventing destabilizing updates. Although IPPO does not explicitly model coordination between agents, it has been shown to achieve competitive performance in several benchmark environments, indicating that decentralized learning with local information can be sufficient in certain settings.

2.2 Multi-Agent Proximal Policy Optimization (MAPPO)

MAPPO [7] extends PPO to multi-agent settings using the Centralized Training with Decentralized Execution (CTDE) paradigm. While policies remain decentralized, MAPPO introduces a centralized critic that leverages additional information during training.

Rather than relying on a simple global value function $V(s)$, the MAPPO critic typically incorporates both global and agent-specific information. In practice, the critic input may consist of the global state augmented with agent-specific features (e.g., local observations or agent identifiers), or a feature-pruned representation that removes redundant components. The value function can therefore be expressed more generally as $V(s, o^1, \dots, o^n, a)$. At execution time, each agent selects actions based only on its local observation $\pi^a(u^a | o^a)$.

By incorporating global information during training, MAPPO reduces the effective non-stationarity perceived by each agent and promotes coordinated behavior. In addition, empirical findings highlight that stabilizing techniques such as value normalization are critical for achieving strong performance, as they reduce scale variance in value targets and improve critic learning.

The relative performance of MAPPO and IPPO depends on the environment. In tasks requiring tight coordination or reasoning about other agents, MAPPO often demonstrates clear advantages due to its centralized critic. However, in settings where local observations are sufficiently informative, IPPO can match or even outperform MAPPO. This suggests that the usefulness of centralized information is inherently task-dependent.

3. Problem Statement

Standard MAPPO [7] relies on a fixed centralized critic throughout training, implicitly assuming that global state information is always the most reliable signal for value estimation. However, our Phase 1 study shows that the benefit of centralized information is task-dependent: MAPPO outperforms IPPO [4] in tightly coupled environments such as Balance and Multiwalker, but IPPO exhibits lower variance and better final performance in Simple World Comm. This suggests that no single fixed critic formulation is universally optimal across cooperative tasks.

We formalize this as the *critic selection problem*: given a cooperative multi-agent task of unknown coordination structure, how should an agent weight global versus local value information during training? Existing methods address this implicitly, wherein MAPPO always conditions the critic on global information, while IPPO relies solely on local observations. Neither adapts to the coordination demands of the environment. We therefore seek a method that subsumes both as special cases and learns the appropriate balance between centralized and decentralized value estimation from data, without requiring manual tuning of the interpolation weight α . Adaptive MAPPO, described in Section 4, is our proposed solution.

4. Method: Adaptive MAPPO

4.1 Motivation: Critic Trade-off

The choice of critic in multi-agent reinforcement learning introduces a fundamental trade-off between bias and variance. As shown in Figure 1, the relative performance of IPPO and MAPPO varies significantly

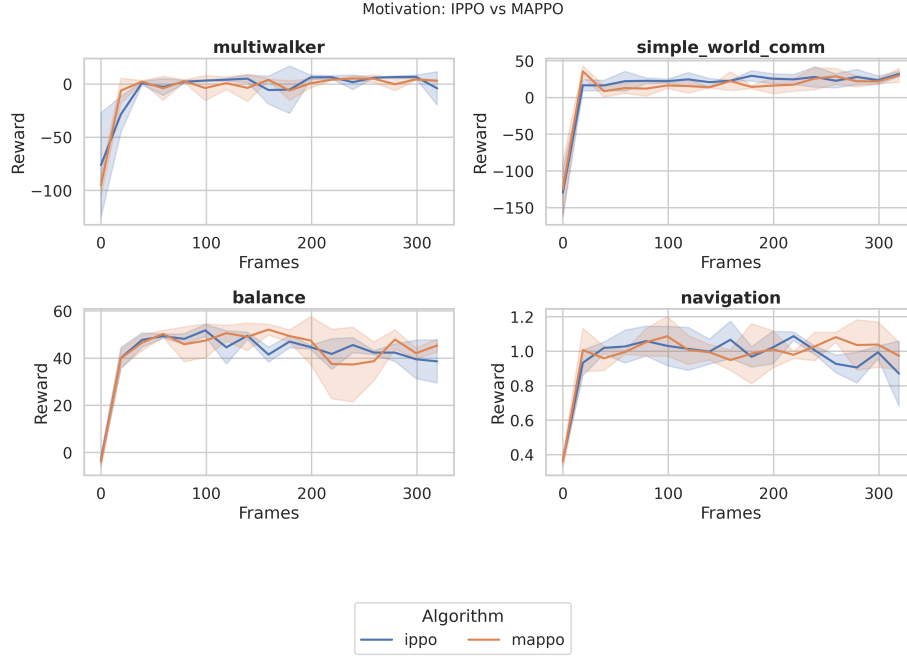


Figure 1: Comparison of IPPO and MAPPO across four environments. The relative performance varies significantly by task, indicating that neither fully decentralized nor fully centralized critic design is universally optimal across environments.

across environments, indicating that neither a fully decentralized nor a fully centralized critic is universally optimal.

A centralized critic, as used in MAPPO, leverages global state information and can therefore provide a more accurate estimate of the true value function. However, this comes at the cost of increased variance due to the higher dimensionality and complexity of the input space. In contrast, a decentralized critic, as used in IPPO, relies only on local observations, resulting in lower variance and more stable learning, but at the expense of bias due to incomplete information about other agents.

This trade-off suggests that critic selection should not be fixed a priori, but instead adapt dynamically depending on the coordination requirements and uncertainty of the environment.

4.2 Mixed Value Function

To combine the advantages of both approaches, we define a mixed value function that interpolates between global and local value estimates:

$$V_{\text{mix}}^a = \alpha^a V_{\text{global}}^a + (1 - \alpha^a) V_{\text{local}}^a \quad (3)$$

where the superscript a denotes the corresponding agent. This formulation reduces to a MAPPO-style critic when $\alpha = 1$ and an IPPO-style critic when $\alpha = 0$, while intermediate values correspond to hybrid critics. As such, it provides a unified framework that continuously spans the spectrum between fully centralized and fully decentralized value estimation.

4.3 Adaptive Gating Mechanism

Rather than using a fixed interpolation coefficient, we model α^a as a learnable function:

$$\alpha^a = \sigma(f_{\theta}(\phi(V_{\text{global}}^a, V_{\text{local}}^a))) \quad (4)$$

where $\phi(\cdot)$ denotes a feature transformation of the value estimates and $\sigma(\cdot)$ ensures that $\alpha^a \in [0, 1]$.

The gating network learns to determine how much to trust global versus local value estimates based on their relationship. When the two estimates differ significantly, the model can favor the signal that better captures coordination effects. When they are consistent, the model can rely more heavily on the lower-variance local estimate. This enables the critic to adapt to different coordination regimes both within and across environments.

4.4 Training Integration

We replace the standard value function in PPO with the mixed value V_{mix} . Advantages are computed using V_{mix} via Generalized Advantage Estimation, while the policy update remains unchanged. This preserves the stability properties of PPO while providing a more flexible and informative value signal.

5. Implementation

5.1 Architecture Overview

The Adaptive MAPPO architecture consists of a local critic, a global critic, and a gating network that combines their outputs. The local critic takes the agent-specific observation o^a as input, while the global critic operates on the global state or a concatenation of observations and agent identifiers. The outputs of both critics are passed to the gating network, which produces the interpolation coefficient α used to compute the mixed value V_{mix} . The gating network is integrated within the critic pipeline and jointly executed with the global and local value networks.

5.2 Critic Networks

The local critic is conditioned only on the agent’s observation, whereas the global critic has access to global state information. Both critics are implemented as multi-layer perceptrons with two hidden layers of size 256 and ReLU activations. This symmetric design ensures comparable capacity while allowing the two critics to capture different aspects of the environment.

5.3 Gating Network

The gating network is a lightweight multi-layer perceptron that predicts the interpolation coefficient α . Instead of operating on raw observations, it uses value-level features derived from the outputs of the critics. These include normalized global and local value estimates, their absolute difference, statistics computed across agents such as variance and deviation from the mean, and a logarithmic ratio capturing relative scale.

We normalize critic outputs using batch-level statistics computed across agents to ensure a shared scale between global and local value estimates. This design constrains the gating mechanism to operate on learned value representations rather than high-dimensional raw inputs, reducing complexity and improving stability.

5.3.1 Mathematical Form of the Gating Network

Let V_g^a and V_ℓ^a denote the global and local critic outputs for agent a . Instead of choosing a fixed interpolation weight, Adaptive-MAPPO computes a learned mixing coefficient $\alpha^a \in [0, 1]$ using a lightweight neural network. First, we normalize the critic outputs using the global critic statistics:

$$\mu_g = \text{mean}(V_g), \quad \sigma_g = \text{std}(V_g) + \varepsilon, \quad (5)$$

$$\tilde{V}_g^a = \frac{V_g^a - \mu_g}{\sigma_g}, \quad \tilde{V}_\ell^a = \frac{V_\ell^a - \mu_g}{\sigma_g}. \quad (6)$$

From these normalized values, we construct the following feature vector:

$$x^a = \left[\tilde{V}_g^a, \tilde{V}_\ell^a, |\tilde{V}_g^a - \tilde{V}_\ell^a|, \text{Var}_j(\tilde{V}_\ell^j), \tilde{V}_\ell^a - \frac{1}{N} \sum_{j=1}^N \tilde{V}_\ell^j, \log \frac{|V_g^a| + \varepsilon}{|V_\ell^a| + \varepsilon} \right]. \quad (7)$$

These features capture: (i) the normalized global estimate; (ii) the normalized local estimate; (iii) their disagreement; (iv) the variance of local values across agents; (v) the deviation of the current agent from the team mean; (vi) the relative scale of the two critics.

The gating network is a one-hidden-layer MLP:

$$h^a = \text{ReLU}(\text{LN}(W_1 x^a + b_1)), \quad (8)$$

$$z^a = W_2 h^a + b_2, \quad \alpha^a = \sigma(z^a), \quad (9)$$

where $\text{LN}(\cdot)$ denotes LayerNorm and $\sigma(\cdot)$ is the sigmoid function.

The final mixed critic is then

$$V_{\text{mix}}^a = \alpha^a V_g^a + (1 - \alpha^a) V_\ell^a. \quad (10)$$

Thus, $\alpha^a \approx 1$ makes the method behave like MAPPO, while $\alpha^a \approx 0$ recovers IPPO. Intermediate values allow the algorithm to adaptively trade off coordination-aware global information and lower-variance local estimation.

5.3.2 Integration with PPO

The mixed value function replaces the standard critic when computing the temporal-difference residual:

$$\delta_t^a = r_t^a + \gamma V_{\text{mix}}^a(s_{t+1}) - V_{\text{mix}}^a(s_t), \quad (11)$$

where V_{mix} is computed using the corresponding α^a at each state.

Generalized Advantage Estimation (GAE) is then computed as

$$\hat{A}_t^a = \sum_{l=0}^{\infty} (\gamma \lambda)^l \delta_{t+l}^a. \quad (12)$$

These advantages are used in the standard PPO clipped objective, while the critic loss is computed using V_{mix}^a . Since V_{mix}^a depends on α^a , gradients from the critic loss also train the gating network end-to-end. The mixed value function is fully differentiable, allowing gradients to flow through both critics into the gating network.

5.4 Training Procedure

For each training batch, we first compute the global and local value estimates. These are then passed through the gating network to obtain α , which is used to construct the mixed value function. Advantages are computed using V_{mix} , and the actor, both critics, and the gating network are updated jointly. The gating network receives gradients through the value loss, enabling it to learn the appropriate interpolation strategy directly from the training signal.

5.5 Loss Function

The total loss is defined as

$$L_{\text{total}} = L_{\text{actor}} + c_1 L_{\text{value}}(V_{\text{mix}}) + c_2 L_{\text{entropy}} \quad (13)$$

Gradients propagate through both critics and the gating network via the mixed value function, allowing all components to be trained end-to-end.

5.6 Integration with BenchMARL

Adaptive MAPPO is implemented within the BenchMARL framework as a modification of the standard MAPPO pipeline. The centralized value estimate is replaced with the mixed value function, and a gating module is introduced to combine the outputs of the global and local critics. This modification introduces

minimal computational overhead while enabling significantly greater flexibility in value estimation.

6. Environments

We evaluate our methods on four cooperative multi-agent tasks spanning two environment families: **VMAS** and **PettingZoo**. These tasks were selected to cover different coordination structures, ranging from loose spatial coordination to tightly coupled physical control and asymmetric communication. This diversity is particularly useful for our study, since Adaptive MAPPO is designed to interpolate between centralized and decentralized value estimation depending on how much global coordination a task requires.

6.1 VMAS Navigation

VMAS Navigation is a cooperative continuous-control task in which three agents must move to their respective targets in a shared 2D world while avoiding interference with one another. In our setup, collisions are enabled, and each episode has a fixed horizon of 100 steps. The reward reflects coordinated progress toward the targets, so good performance requires both accurate individual motion control and awareness of nearby agents. This task is relatively loosely coupled: agents share space, but each one has its own destination. It therefore provides a useful setting for studying whether local information is already sufficient or whether centralized value information still improves learning.

6.2 VMAS Balance

VMAS Balance is a more tightly coupled cooperative control task in which four agents must jointly stabilize and transport a shared object over a horizon of 100 steps. Unlike Navigation, the agents are directly coupled through a common physical objective, so the effect of one agent’s action depends strongly on the actions of the others. The reward, therefore, reflects collective control quality rather than mostly independent progress. This makes Balance a strong benchmark for testing whether methods that rely more heavily on centralized information achieve better stability and coordination.

6.3 PettingZoo Multiwalker

PettingZoo Multiwalker is a cooperative locomotion task with three walkers that must carry a package forward without collapsing. Episodes can last up to 500 steps. The reward includes forward-progress incentives together with instability penalties: in our configuration, a fall incurs a penalty of -10 , and if the fall ends the episode, an additional termination penalty of -100 is applied. We also enable `terminate_on_fall` and `remove_on_fall`, making failure events especially costly. As a result, this environment emphasizes long-horizon coordination, robustness, and stable team behavior during training.

6.4 PettingZoo Simple World Comm

PettingZoo Simple World Comm is a cooperative task with asymmetric observations and explicit communication. A designated leader agent has access to richer information, while the remaining agents must act from limited local observations and coordinate using the shared communication signal. Performance depends on how effectively the team uses the communicated information to reach useful targets while avoiding distractors. This task is especially relevant for our study because it tests whether adaptive mixing between local and global critics can better handle environments where coordination depends on selectively exploiting shared information rather than always relying on a fully centralized view.

7. Experiment Setup

Phase 2 evaluates six PPO-based methods on the same four tasks from Phase 1 using three seeds per configuration: $4 \text{ environments} \times 6 \text{ algorithms} \times 3 \text{ seeds} = 72 \text{ runs}$. The methods are IPPO, MAPPO, fixed hybrid critics with a constant global interpolation coefficient $\alpha \in \{0.25, 0.5, 0.75\}$ applied uniformly across all agents and states, and Adaptive-MAPPO. This setup compares the two endpoint critics, decentralized IPPO and centralized MAPPO, against both manually chosen and learned interpolation strategies. All runs use the same BenchMARL pipeline and core hyperparameters as Phase 1: 2M training frames, learning

Algo	Best	Final	Drop%	$\bar{\alpha}$	Algo	Best	Final	Drop%	$\bar{\alpha}$
IPPO	9.74(± 0.92)	-4.02(± 15.59)	141.3	—	IPPO	38.38(± 3.54)	32.08(± 3.68)	16.4	—
MAPPO	5.98(± 2.59)	3.03(± 1.14)	49.4	—	MAPPO	39.92(± 0.66)	30.16(± 9.41)	24.5	—
Fixed $\alpha = 0.25$	34.19(± 13.32)	15.98(± 28.69)	53.3	—	Fixed $\alpha = 0.25$	41.13(± 3.50)	40.07(± 4.85)	2.6	—
Fixed $\alpha = 0.50$	16.48(± 7.93)	-1.68(± 15.82)	110.2	—	Fixed $\alpha = 0.50$	39.91(± 1.74)	25.82(± 7.36)	35.3	—
Fixed $\alpha = 0.75$	24.34(± 20.73)	5.27(± 35.12)	78.4	—	Fixed $\alpha = 0.75$	38.13(± 10.14)	25.42(± 4.03)	33.3	—
Adaptive	12.48(± 9.62)	9.83(± 11.74)	21.2	0.94(± 0.01)	Adaptive	30.68(± 5.88)	17.48(± 5.71)	43.0	0.37(± 0.07)

(a) Multiwalker					(b) Simple World Comm				
Algo	Best	Final	Drop%	$\bar{\alpha}$	Algo	Best	Final	Drop%	$\bar{\alpha}$
IPPO	52.08(± 2.05)	38.65(± 9.13)	25.8	—	IPPO	1.14(± 0.02)	0.87(± 0.19)	23.7	—
MAPPO	54.06(± 2.38)	45.37(± 2.79)	16.1	—	MAPPO	1.17(± 0.04)	0.97(± 0.08)	16.8	—
Fixed $\alpha = 0.25$	55.39(± 3.28)	47.71(± 4.15)	13.9	—	Fixed $\alpha = 0.25$	1.13(± 0.01)	0.90(± 0.11)	20.3	—
Fixed $\alpha = 0.50$	54.78(± 3.13)	39.81(± 8.34)	27.3	—	Fixed $\alpha = 0.50$	1.12(± 0.03)	0.94(± 0.06)	16.5	—
Fixed $\alpha = 0.75$	53.61(± 1.27)	39.06(± 5.00)	27.1	—	Fixed $\alpha = 0.75$	1.16(± 0.04)	0.96(± 0.06)	17.0	—
Adaptive	53.85(± 1.54)	44.56(± 6.73)	17.2	1.00(± 0.00)	Adaptive	1.12(± 0.05)	0.95(± 0.04)	15.2	0.24(± 0.03)

(c) Balance					(d) Navigation				
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Table 1: Aggregate results (3 seeds each). **Best** = peak mean $\pm$ std; **Final** = final performance mean $\pm$ std; **Drop** = relative decrease from peak to final. $\bar{\alpha}$ denotes the mean mixing coefficient at the end of training (Adaptive only; MAPPO $\equiv \alpha = 1$, IPPO $\equiv \alpha = 0$). **Bold** indicates best Final return and lowest Drop% per environment.

rate 3×10^{-4} , entropy coefficient 0.01, 6000 frames per training iteration, evaluation every 120k frames over 10 episodes, and MLP actor/critic networks with two hidden layers of size 256. Experiments were run on an **NVIDIA A40** GPU node. This controlled setup keeps the training budget, architecture, and evaluation protocol fixed, so performance differences mainly reflect the critic design.

For fixed hybrids, the mixed critic uses a constant coefficient ($\alpha \in \{0.25, 0.5, 0.75\}$). Adaptive-MAPPO instead learns an agent-specific coefficient $\alpha^a \in [0, 1]$ from global/local value features, making the global-local tradeoff data-dependent rather than manually fixed.

8. Results

We report results across the four environments listed in Section 7. The shared protocol uses three seeds per configuration, following the 72-run design. For each run we use evaluation returns, with the *peak* defined as the best evaluation over the 2M-frame trajectory and the *final* defined as the mean over the last 10% of evaluations. All numbers are mean $\pm$ std across seeds.

8.1 Training Dynamics Across Algorithms

Figure 2 presents evaluation-return learning curves for all six algorithms, including IPPO, MAPPO, three fixed- α hybrids, and Adaptive-MAPPO. The curves correspond directly to the aggregated statistics reported in Table 1, which summarize peak performance, final performance, and late-training degradation across seeds. Two consistent patterns emerge across environments.

- Environment-dependent value of centralized information.** The relative advantage of MAPPO over IPPO varies significantly across tasks. In Multiwalker, IPPO shows substantially higher peak performance but suffers from strong degradation, while MAPPO remains comparatively stable but lower-performing overall. In contrast, on Simple World Comm., Balance, and Navigation, the two baselines are closely matched, with differences largely within seed variance (Table 1). This indicates that the benefit of centralized value information is highly task-specific rather than uniform across cooperative settings.
- Non-monotonic training and late-stage degradation.** Most algorithms exhibit a noticeable decline from peak to final performance in at least one environment, as captured by the Drop% column in Table 1. This effect is most pronounced in Multiwalker, where all methods experience substantial degradation,

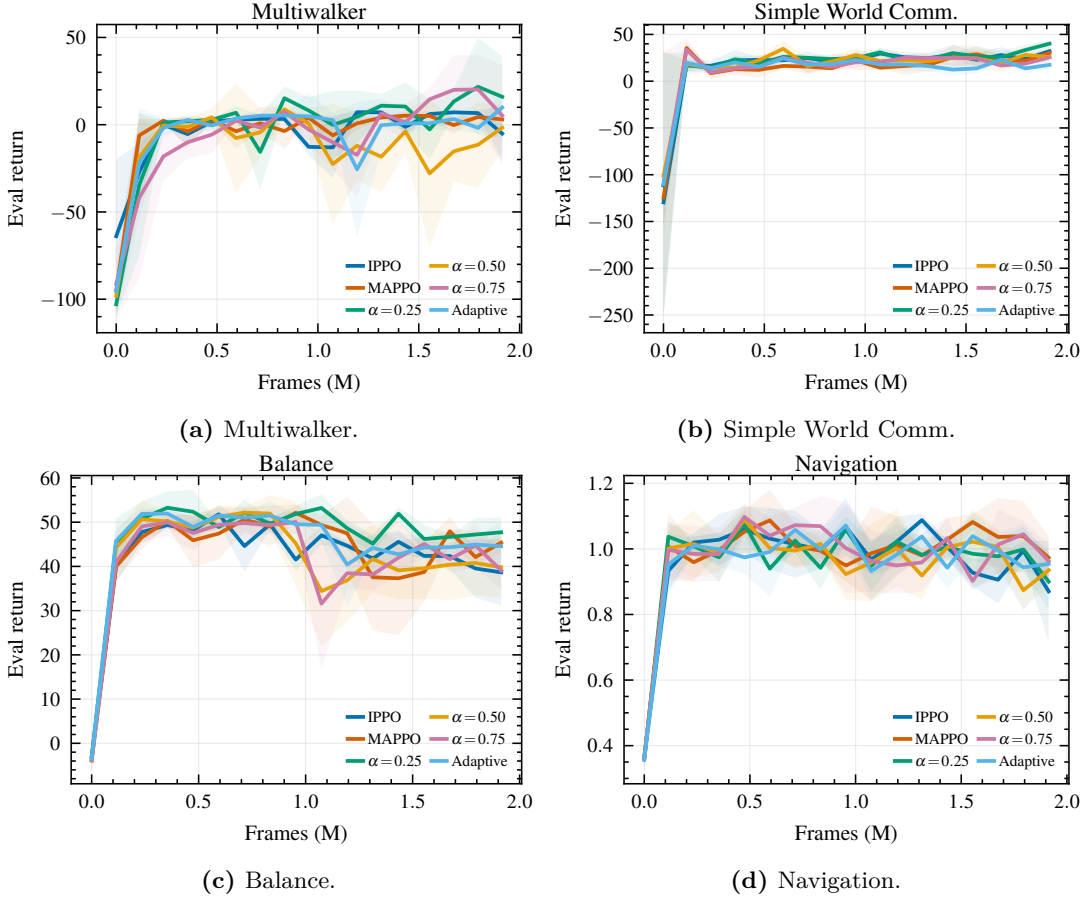


Figure 2: Evaluation-return learning curves for all six algorithms across the four environments (mean $\pm$ std, shaded). Peak and final returns from these curves are reported in Table 1.

whereas Balance and Navigation remain relatively stable. This motivates the need for mechanisms that can adapt the critic structure during training rather than relying on a fixed interpolation strategy.

Overall, these observations highlight that neither fully decentralized nor fully centralized critics are consistently optimal, and that fixed hybrid combinations do not reliably resolve this trade-off across environments.

8.2 Fixed- α Blending

Figure 3 reports both the *peak* return (solid bars) and the *final* return (hatched bars) for all six algorithms. The gap between these values reflects the late-training degradation discussed in Section 8.4. The optimal fixed mixing coefficient is strongly environment-dependent. In Multiwalker, $\alpha = 0.25$ achieves the highest peak but still undergoes substantial late-training degradation (Drop $\approx 53\%$), while intermediate $\alpha = 0.50$ collapses even more severely (Drop $> 110\%$). No single fixed coefficient simultaneously delivers both strong peak performance and stable final returns. In Simple World Comm., the low- α variant ($\alpha = 0.25$) performs best, while larger α values degrade performance. In Balance, all fixed- α variants achieve closely clustered *peaks* around MAPPO ($53.6 \sim 55.4$), although their final returns differ by up to 8.7 units ($39.1 \sim 47.7$). In Navigation, all methods are effectively saturated, with negligible differences across α values. Peak and final performance rankings are not consistent across training. In Multiwalker, $\alpha = 0.25$ achieves the highest peak but undergoes significant degradation by the end of training, whereas higher values such as $\alpha = 0.75$ are more stable and retain a larger fraction of their performance. This indicates that the same fixed interpolation coefficient can favor either peak performance or stability, but not both simultaneously. Overall, these results show that no single fixed α is optimal across environments or across training phases,

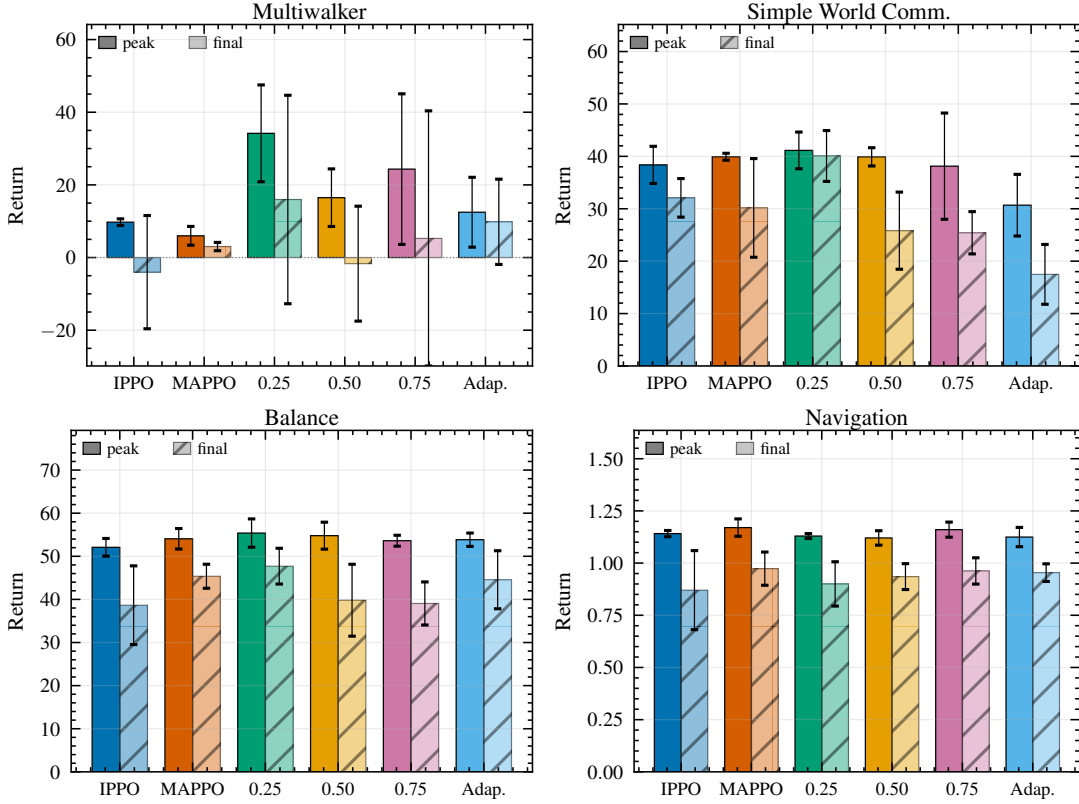


Figure 3: Peak (solid) and final (hatched) evaluation return per environment for all six algorithms: IPPO, MAPPO, fixed- α hybrids, and Adaptive-MAPPO.

directly motivating a learned, state-dependent mixing strategy.

8.3 Adaptive MAPPO

Figure 4a visualises the behaviour of the learned gating coefficient α_t during training. The gating network converges to markedly different operating regimes across environments.

In Multiwalker, the mean α rapidly increases to a high value ($\bar{\alpha} \approx 0.94$), indicating a strong preference for centralized value estimation in tightly coupled coordination tasks. In Balance, α saturates at 1.0, effectively recovering fully centralized critic behavior throughout training. In Simple World Comm., α stabilizes around 0.37, reflecting a mixed regime where both global and local information contribute meaningfully to value estimation under asymmetric observation and communication constraints. In Navigation, α converges to approximately 0.24, indicating a stronger reliance on local value estimates in a setting where decentralized information is largely sufficient. Overall, these results show that the learned mixing coefficient adapts to the coordination structure of the environment, producing distinct operating regimes that cannot be captured by a single fixed interpolation coefficient.

8.4 Training Stability

Peak performance alone can be misleading: a method that briefly achieves high returns but collapses during training is not practically useful. We therefore report the relative performance drop $(\text{peak} - \text{final})/|\text{peak}|$, shown in Figure 4(b). Across environments, IPPO exhibits highly variable stability, with particularly large degradation in Multiwalker, while MAPPO shows moderate decline on Balance, Navigation, and Simple World Comm. (16 ~ 25%), but substantially larger degradation on Multiwalker (49.4%). The fixed- α variants exhibit intermediate behavior, with some configurations (e.g., $\alpha = 0.50$ in Multiwalker) showing severe instability, consistent with the late-training collapse observed in the learning curves. Adaptive

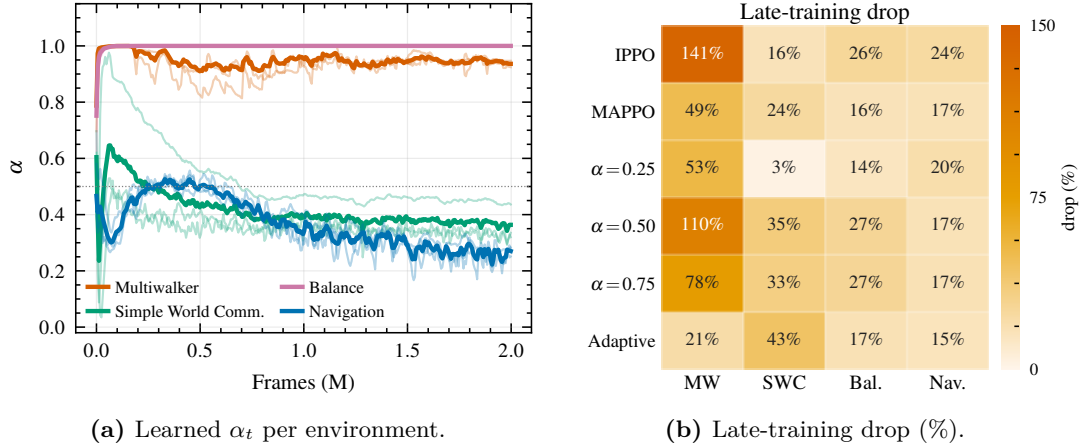


Figure 4: (a) Adaptive MAPPO’s learned mixing coefficient α_t during training, shown across all four environments (thin traces: individual seeds; thick lines: means). The steady-state value of α varies significantly across environments, reflecting different coordination regimes. (b) Relative late-training drop (peak – final)/|peak| per algorithm $\times$ environment; lighter values indicate greater stability.

MAPPO achieves the lowest degradation in Multiwalker and Navigation, and ranks third (17.2%) in Balance within a tight cluster with the two strongest fixed variants (Fixed $\alpha = 0.25$: 13.9%; MAPPO: 16.1%), indicating improved robustness of the learned mixing strategy in 3 of 4 environments. Its stability pattern aligns with the environment-dependent α trajectories shown in Figure 4(a), where the gating mechanism adapts to different coordination regimes rather than remaining fixed.

9. Discussion

Across environments, a consistent pattern emerges regarding when centralized information is beneficial in multi-agent reinforcement learning.

The utility of centralized critics is strongly environment-dependent. In highly coupled tasks such as Multiwalker and Balance, Adaptive MAPPO learns to weight global information heavily ($\bar{\alpha} \geq 0.94$), consistent with the intuition that coordination benefits from centralized estimates; interestingly, the strongest final returns in these two tasks are achieved by the fixed low- α hybrid ($\alpha = 0.25$), suggesting that a small centralized contribution is already sufficient to capture most of the coordination benefit. In weaker coupling regimes such as Navigation, decentralized information is largely sufficient and centralized value estimation provides only marginal gains.

At the same time, no fixed interpolation coefficient α is optimal across all environments or throughout training (Section 8.2). This rules out any universal hand-tuned mixing strategy between centralized and decentralized critics, as different environments and training phases require different bias–variance trade-offs.

Adaptive MAPPO addresses this limitation by learning environment-specific mixing behavior. It achieves the lowest or near-lowest relative performance drop in 3 out of 4 environments, indicating improved robustness compared to fixed interpolation strategies. However, in Simple World Comm., it substantially under-performs the best fixed- α variant (Drop 43.0% vs. 2.6% for $\alpha = 0.25$), suggesting that the learned gating mechanism is less effective in settings with asymmetric or partially communicative structure.

These results suggest that adaptive critic mixing provides a more flexible alternative to fixed centralized or decentralized critic designs.

Results are based on three seeds per configuration, so reported trends should be interpreted as indicative rather than statistically conclusive. The evaluation is limited to cooperative or weakly mixed-cooperative environments from BenchMARL. Finally, the gating mechanism relies on engineered features derived from critic outputs, and learning these representations directly from raw observations remains an open direction.

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